

DKW

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1 High Level Notes

[CLICK] simulations of method

1.1 Roberto Imbuzeiro Oliveira lecture notes

- This proof is from Roberto Imbuzeiro Oliveira lecture notes.
- Notes from Robert Imbuzeiro Oliveira give a *weak* DKW inequality.
- Tight bound is $2 \exp(-\varepsilon^2 2n)$, can be proven using VC and non-VC methods. Lecture notes give $4 \exp(-n\varepsilon^2/8)$. Much weaker C and constant inside $\exp()$.
- I think we can get tight bound, just need more time to prove it.
- If we can't then the weak bound isn't very good unless n is big $n > 2000$, which can be useful for big RCT.

1.2 Miscellaneous

- Good amount of literature on DKW for sampling with or without replacement.
- Had to cheat and use $\frac{1}{pn}$ instead of $1/N_1$ because DKW doesn't play nice with dividing by random N_1 .
- Proofs seem to translate well, I think I must be more rigorous about Lemma 3.2, namely how to do the ε bound

2 Proving DKW for ind but not identical rv

2.1 Step 1

Prove that

$$\mathbb{E} \left[\exp \left(\theta \sup_{0 \leq i \leq n} S_i \right) \right] \leq 2 \exp(\theta^2 n / 2)$$

where S_i is the running sum of a random walk on the integers and $S_0 = 0$.

Zeroth part:

$$\Pr(S_i \geq 0) \geq \frac{1}{2}$$

(equally likely to be both above and below and some mass at 0).

First part use reflection principle to say

$$\Pr\left(\sup_{0 \leq i \leq n} S_j \geq t\right) \leq 2 \Pr(S_n \geq t)$$

Second part: for f diffble, prove

$$\mathbb{E}[f(X); X \geq 0] = f(0) \Pr(X \geq 0) + \int_0^\infty f'(t) \Pr(X \geq t) dt.$$

Third part: let $f(t) = \exp(\theta t)$ with $f(\sup_{0 \leq i \leq n} S_i) = \exp(\theta \sup_{0 \leq i \leq n} S_i)$.

$$\begin{aligned} \mathbb{E}[f(X)] &= \mathbb{E}[f(X); X \geq 0] = \underbrace{f(0) \Pr\left(\sup_{0 \leq i \leq n} S_i \geq 0\right)}_{=1} + \int_0^\infty \theta \exp(\theta t) \Pr\left(\sup_{0 \leq i \leq n} S_i \geq t\right) dt \\ &\leq 1 + 2\theta \int_0^\infty \exp(\theta t) \Pr(S_n \geq t) dt \\ &= 1 + 2\mathbb{E}[(\exp(\theta S_n) - 1) \mathbb{I}_{S_n \geq 0}] \text{ Layer Cake} \\ &= 1 + \underbrace{2\mathbb{E}[\exp(\theta S_n) \mathbb{I}_{S_n \geq 0}]}_{\leq 2\mathbb{E}[\exp(\theta S_n)]} - 2 \Pr(S_n \geq 0) \\ &\leq 2\mathbb{E}[\exp(\theta S_n)] \leq 2 \exp(\theta^2 n/2), \end{aligned}$$

where the last line comes from MGF.

3 Main Part

Let the potential outcomes be fixed

- $Y_1(1), \dots, Y_n(1)$ have ecdf $F_n^{(1)}$
- $Y_1(0), \dots, Y_n(0)$ have ecdf $F_n^{(0)}$
- $Z_i \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(p)$.

Define our estimate of $F_n^{(1)}$ to be

$$\hat{F}_n^{(1)}(x) := \frac{1}{n \times p} \sum_{i=1}^n \mathbb{I}_{(-\infty, x]} Z_i Y_i(1).$$

This is not the usual definition of an ECDF. We must use the deterministic $n \times p$, because the proof doesn't play nice with random N_1 . We'll prove for $F_n^{(1)}$, but can do for control, too with $n \times (1 - p)$. Drop the (1) for notation

3.1 Dealing with Design Based Inference

The usual DKW bound is not meant for randomness only through treatment, but it can be adapted for it. We have to convince ourselves of a few things before we begin the proofs.

Lemma 3.1 (Symmetrization Lemma). *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}_+ \cup \infty$ be convex. If $X_1, \dots, X_n$ are independent on $\mathcal{X}$ and F is a countable family of measurable function from $\mathcal{X} \rightarrow \mathbb{R}$ with $\mathbb{E}[|f(X_1)|] < \infty$. Then*

$$\mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) \right] \leq \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n f(X_i) \zeta_i \right) \right] \quad (1)$$

For brevity, we call $X_1, \dots, X_n = X$

Lemma 3.2 (Potential Outcome Fix). *Let $F = \{\pm 1 \frac{1}{pn} \mathbb{I}_{(-\infty, x]} : x \in \mathbb{R}_{<0}\}$, $\hat{F}_n(x) = \frac{1}{pn} \sum_{i=1}^n \mathbb{I}_{(-\infty, x]}(Z_i Y_i)$, and $F_n(x) = n^{-1} \sum_{i=1}^n \mathbb{I}_{(-\infty, x]}(Y_i)$, then*

$$\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| = \sup_{f \in F} \left| \sum_{i=1}^n f(Z_i Y_i) - \mathbb{E}[f(Z_i Y_i)] \right|.$$

Now, apply symmetrization lemma with F described in potential outcome fix lemma.

Theorem 3.3 (DKW Theorem).

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq 4 \exp(-\varepsilon^2 (np) / 2).$$

Proof. By Markov's inequality $\forall \theta > 0$,

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq \exp(-\theta \varepsilon) \mathbb{E} \left[\exp \left(\theta \sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| \right) \right]. \quad (2)$$

$$\sup_{x \in \mathbb{R}_{<0}} \left| \hat{F}_n(x) - F_n(x) \right| = \sup_{f \in F} \left| \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right|.$$

Let our convex function be $\varphi(t) = \exp(\theta t)$ for $\theta > 0$.

$$\begin{aligned} \mathbb{E} \left[\exp \left(\theta \sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F(x) \right| \right) \right] &= \mathbb{E} \left[\varphi \left(\sup_{f \in F} \left| \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right| \right) \right] \\ &= \mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) \right], \end{aligned} \quad (3)$$

where equation 3 is because φ is increasing for $\theta > 0$.

Recall the lemma (1).

$$\mathbb{E} \left[\sup_{f \in F} \varphi \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right] \leq \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n f(X_i) \zeta_i \right) \right],$$

which for our specific φ yields

$$\mathbb{E} \left[\sup_{f \in F} \exp \left(\theta \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) \right] \leq \mathbb{E} \left[\sup_{f \in F} \exp(\theta 2 \sum_{i=1}^n f(X_i) \zeta_i) \right].$$

We recall f is of the form

$$f(z) = \frac{1}{pn} b \mathbb{I}_{(-\infty, x]}(z).$$

Thus we translate

$$\begin{aligned} \mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\frac{\theta 2b}{pn} \sum_{i=1}^n \mathbb{I}_{(-\infty, x]}(X_i) \zeta_i \right) \right] &= \mathbb{E} \left[\mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\frac{\theta 2b}{pn} \sum_{i=1}^n \zeta_i \mathbb{I}_{(-\infty, x]}(X_i) \right) \mid X \right] \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\frac{\theta 2b}{pn} \sum_{i=1}^n \zeta_i \mathbb{I}_{(-\infty, x]}(x_i) \right) \mid X = x \right] \right]. \end{aligned} \quad (4)$$

We sort the x so that $x_1 \leq \dots \leq x_n$ and note that

$$\sum_{i=1}^n \zeta_i \mathbb{I}_{(-\infty, x]}(x_i) = \begin{cases} 0 & x_i > x \\ S_i & x_i < x < x_{i+1} \\ S_n & x \geq x_n \end{cases}$$

where S_i are the running sums of the rademacher random variables.

We recall our first lemma and write

$$\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\theta 2b pn^{-1} \sum_{i=1}^n \zeta_i \mathbb{I}_{(-\infty, x]}(x_i) \right) = \sup_{b \in \pm 1, 0 \leq i \leq n} \exp(\theta 2b (pn)^{-1} S_i).$$

To get rid of the b we consider either $b = 1, b = -1$

$$\sup_{b \in \pm 1, 0 \leq i \leq n} \exp(\theta 2b (pn)^{-1} S_i) \leq \sup_{0 \leq i \leq n} \exp(\theta 2 (pn)^{-1} S_i) + \exp(-\theta 2 (pn)^{-1} S_i).$$

Since S_i are symmetric around 0,

$$\sup_{0 \leq i \leq n} \exp(\theta 2 (pn)^{-1} S_i) + \exp(-\theta 2 (pn)^{-1} S_i) = 2 \sup_{0 \leq i \leq n} \exp(\theta 2 (pn)^{-1} S_i). \quad (5)$$

We recall what we're doing in equation (4) and focus on the inner expectation

$$\begin{aligned} \mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp(\theta 2b (pn)^{-1} \sum_{i=1}^n \zeta_i \mathbb{I}_{(-\infty, x]}(x_i) \mid X = x) \right] &\leq 2 \mathbb{E} \left[\sup_{0 \leq i \leq n} \exp(\theta 2 (pn)^{-1} S_i) \right] \\ &= 2 \mathbb{E} \left[\exp \left(2\theta pn^{-1} \sup_{0 \leq i \leq n} S_i \right) \right] \\ &\leq 4 \exp(\theta^2 2 (pn)^{-1}) \end{aligned} \quad (6)$$

where equation (6) comes from our original (0th) lemma.

This is independent of the observed values $X = x$, so **No longer true if N_1 is random.**
This is why we do $1/np$

$$\mathbb{E} \left[\exp \left(\theta \sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F(x) \right| \right) \right] \leq 4 \exp(\theta^2 2(pn)^{-1}) \quad (7)$$

We recall equation 2 is

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq \exp(-\theta \varepsilon) \mathbb{E} \left[\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| \right],$$

which in turn means

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq 4 \exp(\theta^2 2n^{-1} - \theta \varepsilon).$$

This holds for all θ so minimizing the RHS wrt θ by taking a derivative yields $\theta = \frac{pn\varepsilon}{4}$.
Yielding the final result.

Theorem 3.4.

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq 4 \exp(-pn\varepsilon^2/8)$$

□

4 Connection to Design Based Causal Inference

For simplicity, we let n be the total number of units with a BRE. $Z_i \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(1/2)$. Potential outcomes $Y(0), Y(1)$ are fixed and all the randomness is due to Z . $Y = ZY(1) + (1 - Z)Y(0)$ reveals us partial information.

We can consider the true distribution of $Y(1)$ as

$$F_n^{(1)}(x) = \sum_{i=1}^n \mathbb{I}_{(-\infty, x]}(Y_i(1)).$$

Similarly, we have

$$F_n^{(0)}(x) = \sum_{i=1}^n \mathbb{I}_{(-\infty, x]}(Y_i(0)).$$

We can construct estimators for these using

$$\hat{F}_n^{(1)}(x) = (pn)^{-1} \sum_{i=1}^n \mathbb{I}_{(-\infty, x]} ZY_i,$$

and

$$\hat{F}_n^{(2)}(x) = ((1-p)n)^{-1} \sum_{i=1}^n \mathbb{I}_{(-\infty, x]} (1 - Z)Y_i.$$

By the theorem above for a specified α , we can bound uniformly

$$\Pr\left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n^{(i)}(x) - F_n^{(i)}(x) \right| > \varepsilon_\alpha\right) \leq \alpha, i = 0, 1.$$

But we want instead the distribution of the difference

$$G_n(t) \equiv n^{-1} \sum_{i=1}^n \mathbb{I}_{(-\infty, t]} Y_i(1) - Y_i(0)$$

which is never observed but can be bounded using Makarov's (not Markov) inequality [Makarov, 1982, Firpo et al., 2007].

Theorem 4.1 (Makarov Inequality). *This is an a.s. statement. The CDF of the difference will always be between these two bounds.*

$$G_n(t) \in \left(\sup_x F_n^{(1)}(x) - F_n^{(0)}(x - t) \vee 0, \inf_x F_n^{(1)}(x) - F_n^{(0)}(x - t) + 1 \wedge 1 \right)$$

We don't have access to the $F_n^{(i)}$, but have access to estimates of them with uniform confidence bounds. Therefore we can turn this a.s. bound into a with high probability bound.

Lemma 4.2 (Bound on Difference in Marginals).

$$\Pr\left(F_n^{(1)}(x) - F_n^{(0)}(x - t) \in \left(\hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t) - \varepsilon, \hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t) + \varepsilon\right)\right) \geq 1 - \alpha,$$

where $\varepsilon = \varepsilon_1 + \varepsilon_0$ choosing $\varepsilon_1, \varepsilon_2$ to satisfy $\alpha/2$.

Proof.

$$\begin{aligned} & \Pr\left(F_n^{(1)}(x) - F_n^{(0)}(x) \notin \left(\hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t) - \varepsilon, \hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t) + \varepsilon\right)\right) \\ & \stackrel{\text{Union}}{\leq} \Pr\left(F_n^{(1)}(x) \notin \left(\hat{F}_n^{(1)}(x) - \varepsilon, \hat{F}_n^{(1)}(x) + \varepsilon\right)\right) \\ & + \Pr\left(F_n^{(0)}(x) \notin \left(\hat{F}_n^{(0)}(x - t) - \varepsilon, \hat{F}_n^{(0)}(x - t) + \varepsilon\right)\right) \leq \alpha \end{aligned}$$

□

A Proofs

Sym-Lemma. We can check that

$$\sum_{i=1}^n (f(X_i) - \mathbb{E}[f(X_i)]) = \mathbb{E} \left[\sum_{i=1}^n (f(X_i) - f(Y_i)) \mid X \right]$$

Therefore

$$\begin{aligned} \sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) &= \sup_{f \in F} \varphi \left(\mathbb{E} \left[\sum_{i=1}^n f(X_i) - f(Y_i) \mid X \right] \right) \\ &\leq \sup_{f \in F} \mathbb{E} \left[\varphi \left(\sum_{i=1}^n f(X_i) - f(Y_i) \right) \mid X \right] \end{aligned} \quad (8)$$

$$\leq \mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - f(Y_i) \right) \mid X \right], \quad (9)$$

where equation (8) comes from Jensen's inequality and equation (9) comes from sup of expectation is less than the expectation of the sup.

Iterated expectation gives

$$\mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) \right] \leq \mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - f(Y_i) \right) \right]$$

Define

$$\begin{aligned} X_i^+ &= \begin{cases} X_i & \zeta_i = 1 \\ Y_i & \zeta_i = -1 \end{cases} \\ X_i^- &= \begin{cases} Y_i & \zeta_i = 1 \\ X_i & \zeta_i = -1 \end{cases} \end{aligned}$$

This construction implies

$$(X_i^+, X_i^-) \stackrel{d}{=} (X_i, Y_i).$$

Therefore,

$$\mathbb{E} \left[\sup_{f \in F} \sum_{i=1}^n f(X_i) - f(Y_i) \right] = \mathbb{E} \left[\sup_{f \in F} \sum_{i=1}^n f(X_i^+) - f(X_i^-) \right] \quad (10)$$

One can check that

$$\zeta_i (f(X_i) - f(Y_i)) = f(X_i^+) - f(X_i^-) \implies \sum_{i=1}^n \zeta_i (f(X_i) - f(Y_i)) = \sum_{i=1}^n f(X_i^+) - f(X_i^-).$$

Therefore

$$\mathbb{E} \left[\sup_{f \in F} \sum_{i=1}^n f(X_i^+) - f(X_i^-) \right] = \mathbb{E} \left[\sup_{f \in F} \sum_{i=1}^n \zeta_i (f(X_i) - f(Y_i)) \right]. \quad (11)$$

Putting together equation (10) and equation (11) gives

$$\mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - f(Y_i) \right) \right] = \mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n \zeta_i f(X_i) - \zeta_i f(Y_i) \right) \right].$$

We now use the convexity of φ to get

$$\begin{aligned} \varphi \left(\sum_{i=1}^n \zeta_i f(X_i) - \zeta_i f(Y_i) \right) &= \varphi \left(\sum_{i=1}^n \zeta_i f(X_i) + \sum_{i=1}^n -\zeta_i f(Y_i) \right) \\ &= \varphi \left(\frac{1}{2} \left(2 \sum_{i=1}^n \zeta_i f(X_i) + \sum_{i=1}^n -2\zeta_i f(Y_i) \right) \right) \\ &\leq \frac{1}{2} \varphi \left(2 \sum_{i=1}^n \zeta_i f(X_i) \right) + \varphi \left(2 \sum_{i=1}^n -\zeta_i f(Y_i) \right). \end{aligned} \quad (12)$$

Therefore,

$$\begin{aligned} \mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n \zeta_i f(X_i) - \zeta_i f(Y_i) \right) \right] &\leq \frac{1}{2} \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n \zeta_i f(X_i) \right) + \varphi \left(2 \sum_{i=1}^n -\zeta_i f(Y_i) \right) \right] \\ &= \frac{1}{2} \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n \zeta_i f(X_i) \right) \right] + \frac{1}{2} \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n -\zeta_i f(Y_i) \right) \right] \\ &= \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n f(X_i) \zeta_i \right) \right], \end{aligned} \quad (13)$$

where equation (13) comes from the fact $(X_i, \zeta_i) \stackrel{d}{=} (Y_i, -\zeta_i)$. □

References

- Sergio Firpo, PUC Departamento de Economia, and Geert Ridder. Bounding the Distribution of Treatment Effects. 2007.
- GD Makarov. Estimates for the distribution function of a sum of two random variables when the marginal distributions are fixed. *Theory of Probability & its Applications*, 26(4): 803–806, 1982. Publisher: SIAM.